CCSA105

POLICY LAYERS

Practical Assignment Report

Student information	Details
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Course	CCSA105 – Policy Layers (Module 5)
Date	8 November 2026
Score	_____ / 100

Submission status: Configuration evidence compiled; gateway-dependent installation and traffic tests remain outstanding.

Report status and evidence conventions

This report records the SmartConsole configuration completed during the practical sessions and distinguishes it from tests that require an active Check Point Security Gateway. A published policy package is not the same as an installed policy. No installation success or traffic-log result is claimed where the gateway was unavailable.

Status	Meaning
Completed / verified	The setting or rule is visible in the screenshots or was confirmed during the guided configuration.
Pending	The assignment requires an active Security Gateway or live traffic test, which was not available in this lab session.

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- 1. Introduction and lab environment
- 2. Task 1 – Explore the default policy and layer
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1. Introduction and lab environment

The assignment uses the Orion Logistics scenario. The objective is to divide the Access Control policy into two Ordered Layers—Network and Service-Control—and attach a separate Inline Layer, DMZ-Policy, to the Internal to DMZ rule. The practical also examines how ordered evaluation, cleanup actions and layer sharing affect policy management.

Component	Lab detail
Management	A-SMS (Security Management Server)
Administration workstation	Windows 10 x64 virtual machine running SmartConsole
Policy package	Orion-Layers
Ordered layers	Network; Service-Control
Inline layer	DMZ-Policy, attached to Network Rule 3
Limitation	Security Gateway VM could not be run because of host hardware/resource constraints.

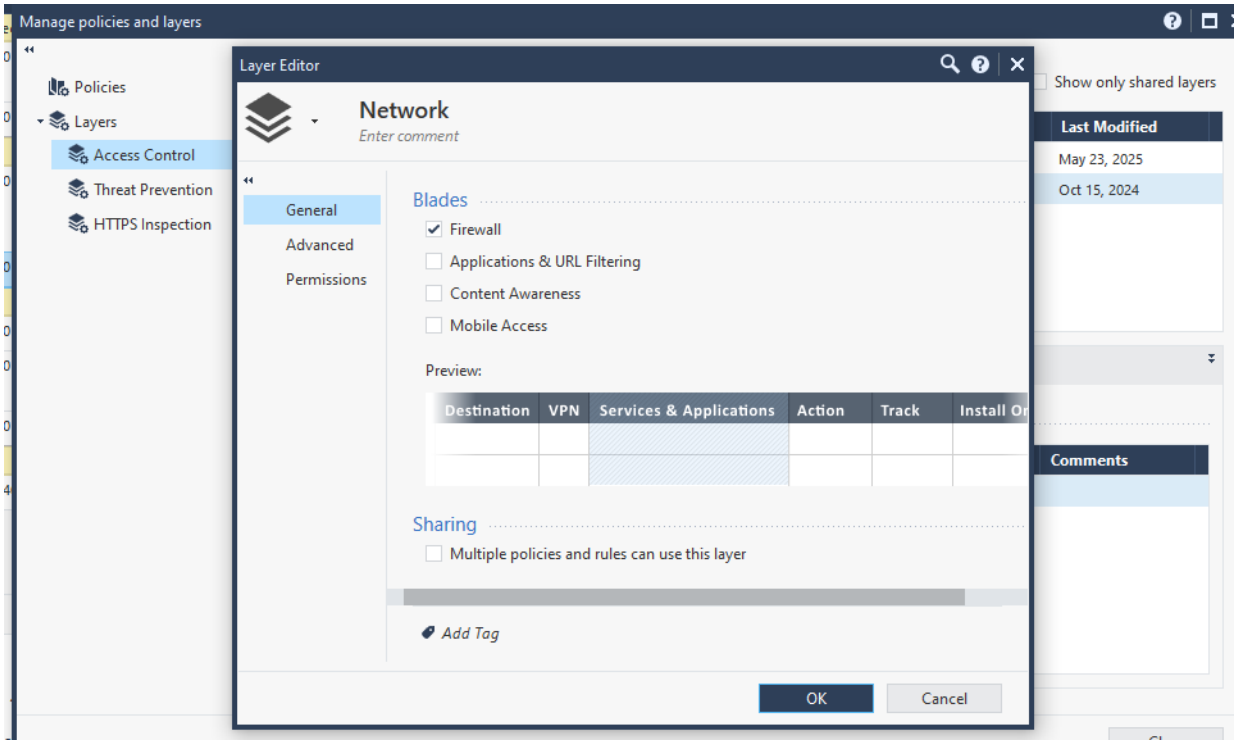
The workstation and management server were observed on the VMware NAT network. The Admin_PC object was corrected to represent the Windows 10 workstation address (10.1.1.128). The assignment's example Admin_PC address (10.1.1.101) was not used because the lab uses its own addressing.

2. Task 1 – Explore the default policy and layer

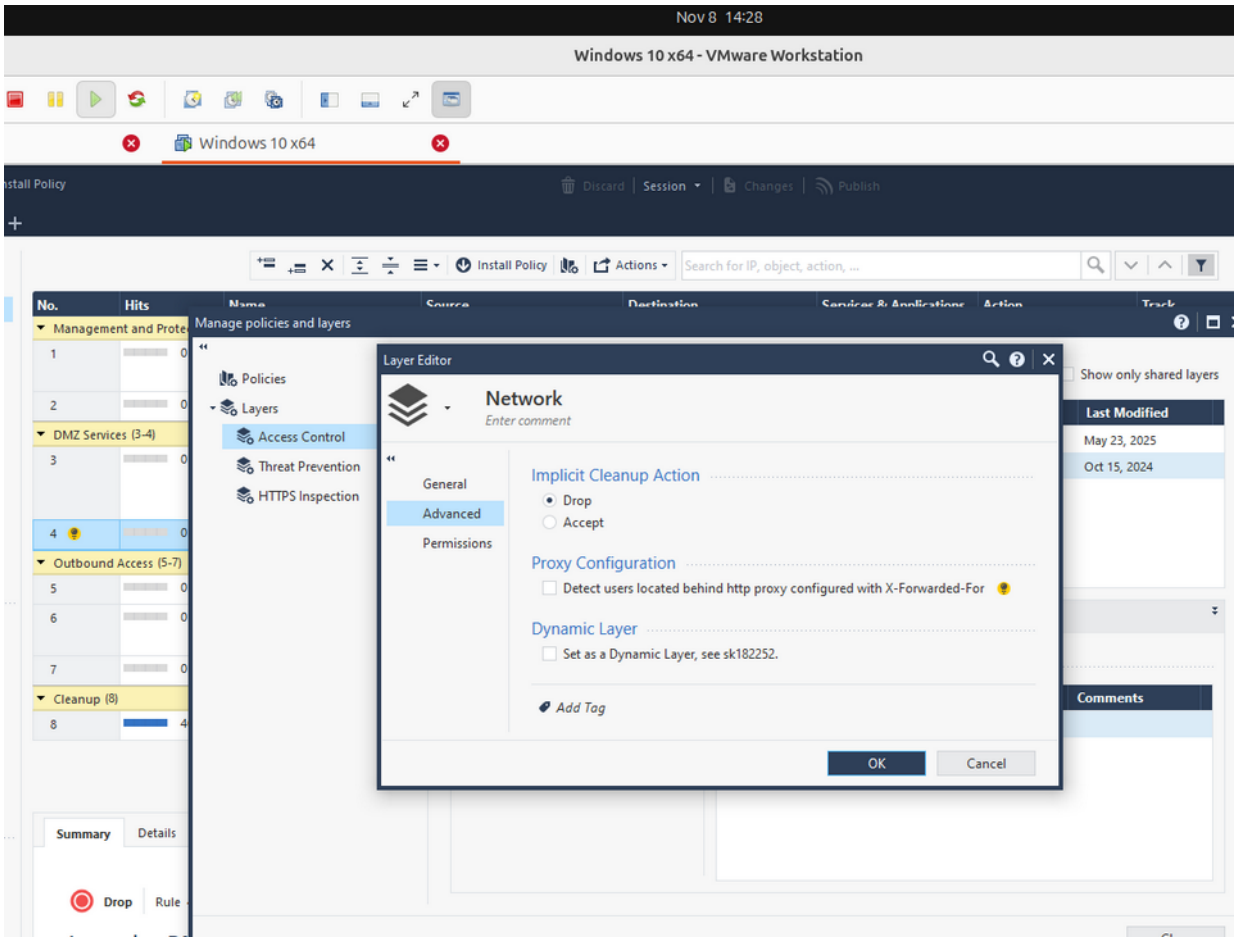
The default Access Control package shown in Manage Policies and Layers is Standard, and its default Ordered Layer is Network. The inspected Network layer showed Firewall enabled, Sharing disabled, Implicit Cleanup Action set to Drop, Proxy Configuration disabled, Dynamic Layer disabled, and the Super User and Read Write All profiles able to edit.

Setting group	Observed value
Default policy package	Standard
Default Ordered Layer	Network
General – blades	Firewall enabled; Applications & URL Filtering, Content Awareness and Mobile Access disabled
General – Sharing	Disabled
Advanced – Implicit Cleanup Action	Drop
Advanced – Proxy Configuration	Disabled
Advanced – Dynamic Layer	Disabled
Permissions	Super User and Read Write All

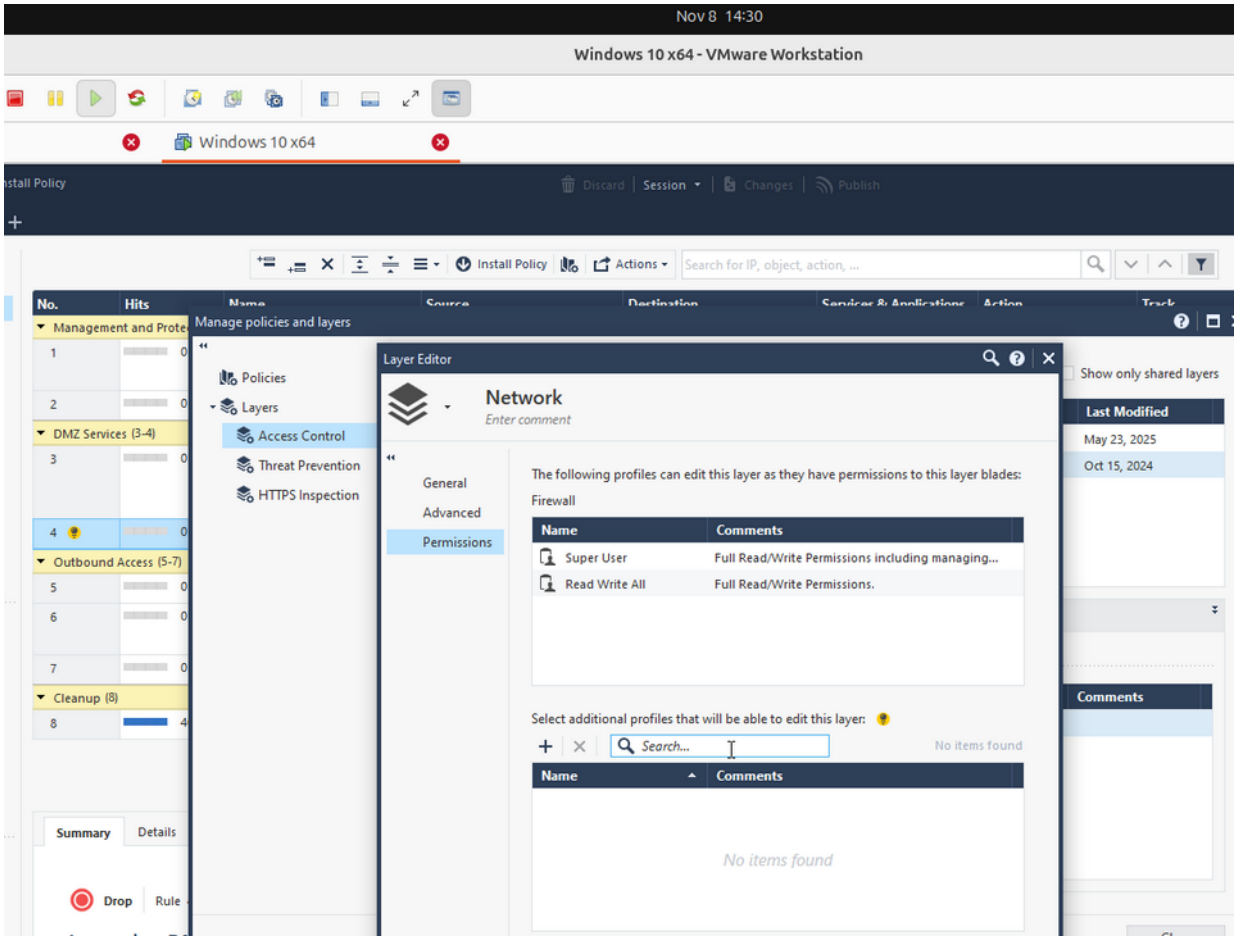
The default rulebase cleanup action was inspected as part of the task. The captured screenshots below show the layer settings pages; include the default cleanup-rule screenshot if it is not already present in the final evidence folder.



Screenshot T1-A: General settings of the inspected Network layer.



Screenshot T1-B: Advanced settings showing Implicit Cleanup Action Drop.

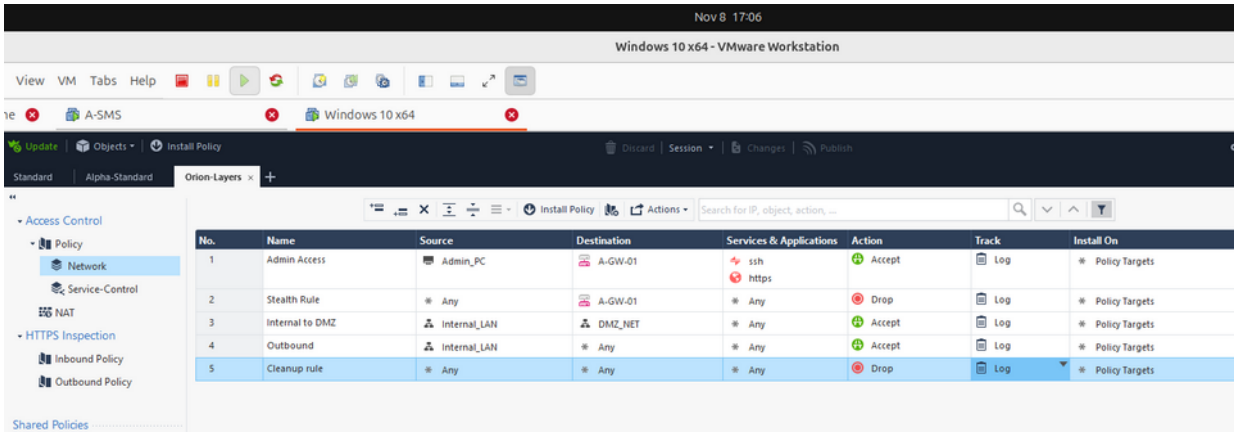


Screenshot T1-C: Permissions showing Super User and Read Write All.

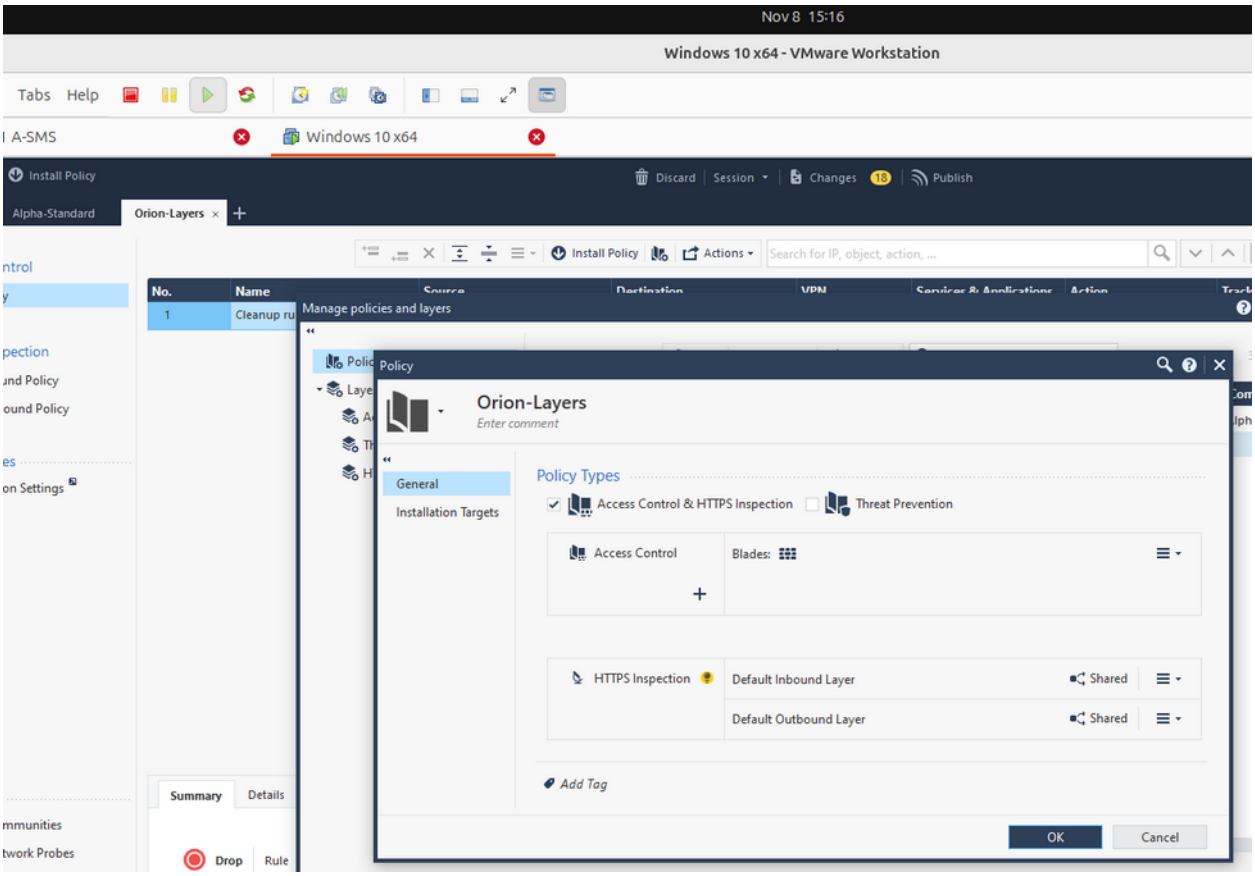
3. Task 2 – Create Orion-Layers and Network

The Orion-Layers Access Control policy package was created. Its Network Ordered Layer was configured with five rules in the required order. Rule 3 was left as Accept at this stage, as directed by the assignment, and was changed to an Inline Layer in Task 5.

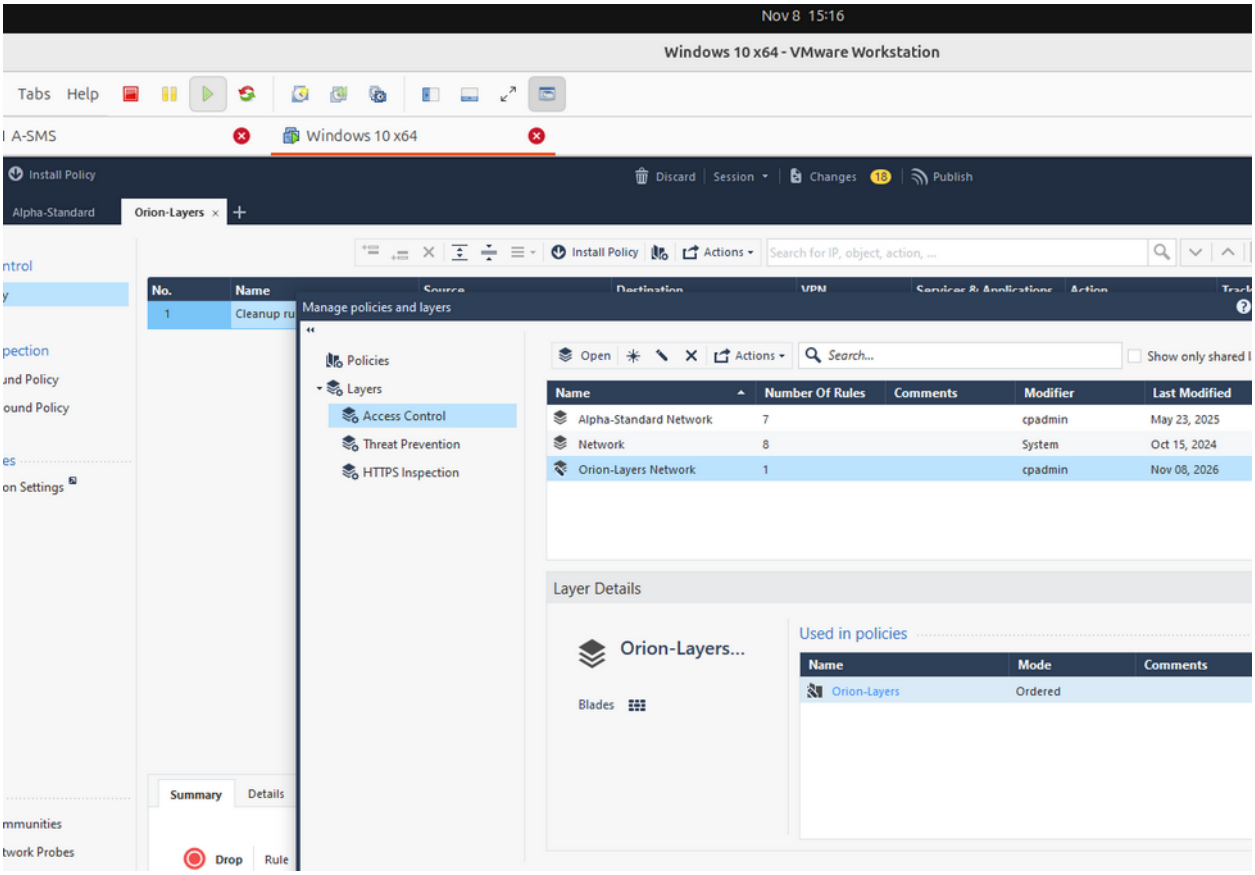
No.	Name	Source	Destination	Services	Action	Track
1	Admin Access	Admin_PC	A-GW-01	SSH, HTTPS	Accept	Log
2	Stealth Rule	Any	A-GW-01	Any	Drop	Log
3	Internal to DMZ	Internal_LAN	DMZ_NET	Any	Accept (initially)	Log
4	Outbound	Internal_LAN	Any	Any	Accept	Log
5	Cleanup Rule	Any	Any	Any	Drop	Log



Screenshot T2-A: Network rulebase showing the five configured rules and Log tracking.



Creation of the Orion-Layers policy package with Access Control enabled and Threat Prevention disabled.



Configuration of the Network Ordered Layer within the Orion-Layers policy package.

4. Task 3 – Add Service-Control

A second Ordered Layer named Service-Control was added below Network in Orion-Layers. The layer was configured with Firewall enabled, Implicit Cleanup Action Drop, Proxy Configuration disabled, Dynamic Layer disabled, and edit permissions for Super User and Read Write All.

The assignment specifies Sharing Disabled for Service-Control. During configuration, the layer appeared as Shared in the package UI after sharing had been enabled to make it selectable. This is a configuration discrepancy to review in SmartConsole before final submission; do not describe its sharing state as Disabled unless it is verified.

No.	Name	Source	Destination	Services	Action	Track
1	Allowed Services	Any	Any	HTTP, HTTPS, DNS (UDP/TCP), SMTP	Accept	Log
2	Cleanup Rule	Any	Any	Any	Drop	Log

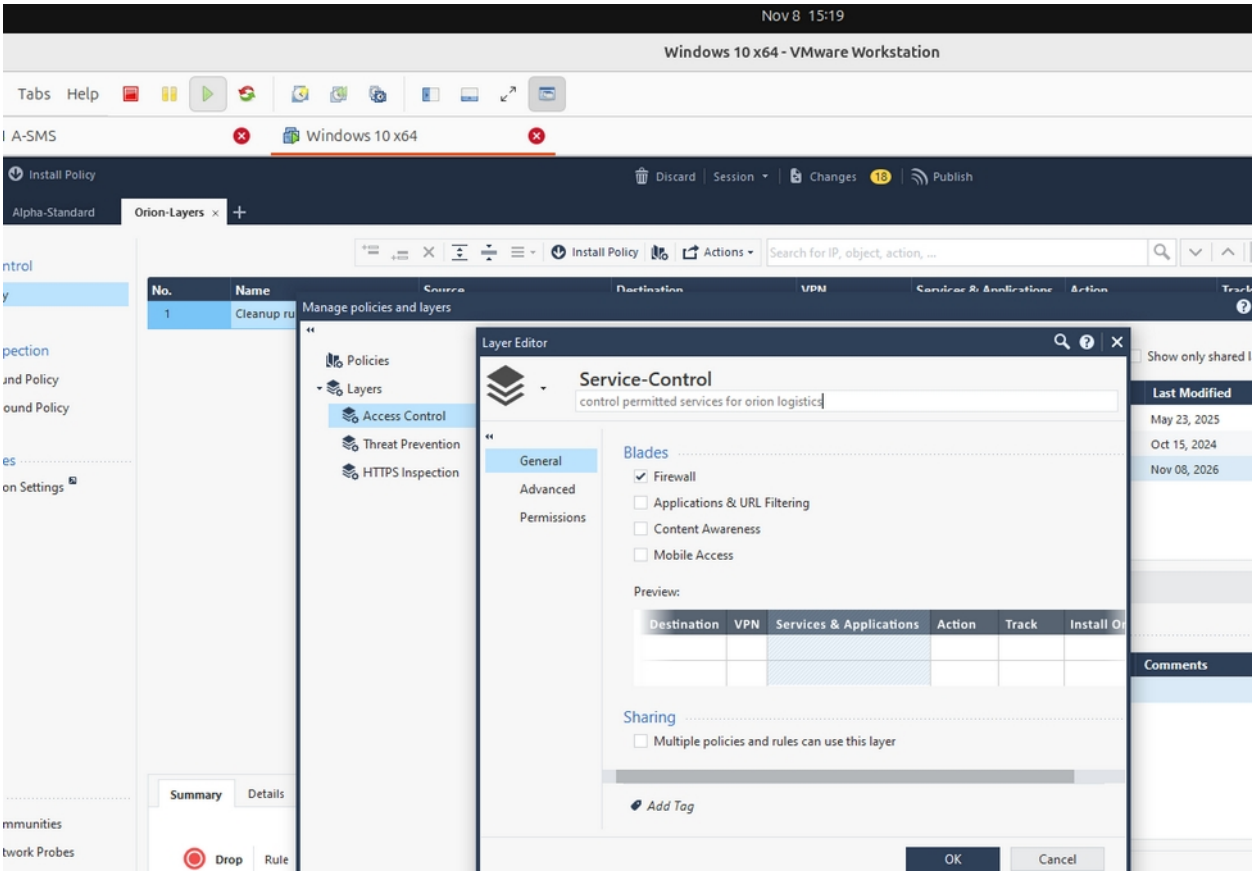


Figure C1: General configuration of the Service-Control layer, with the Firewall blade enabled and Sharing disabled.

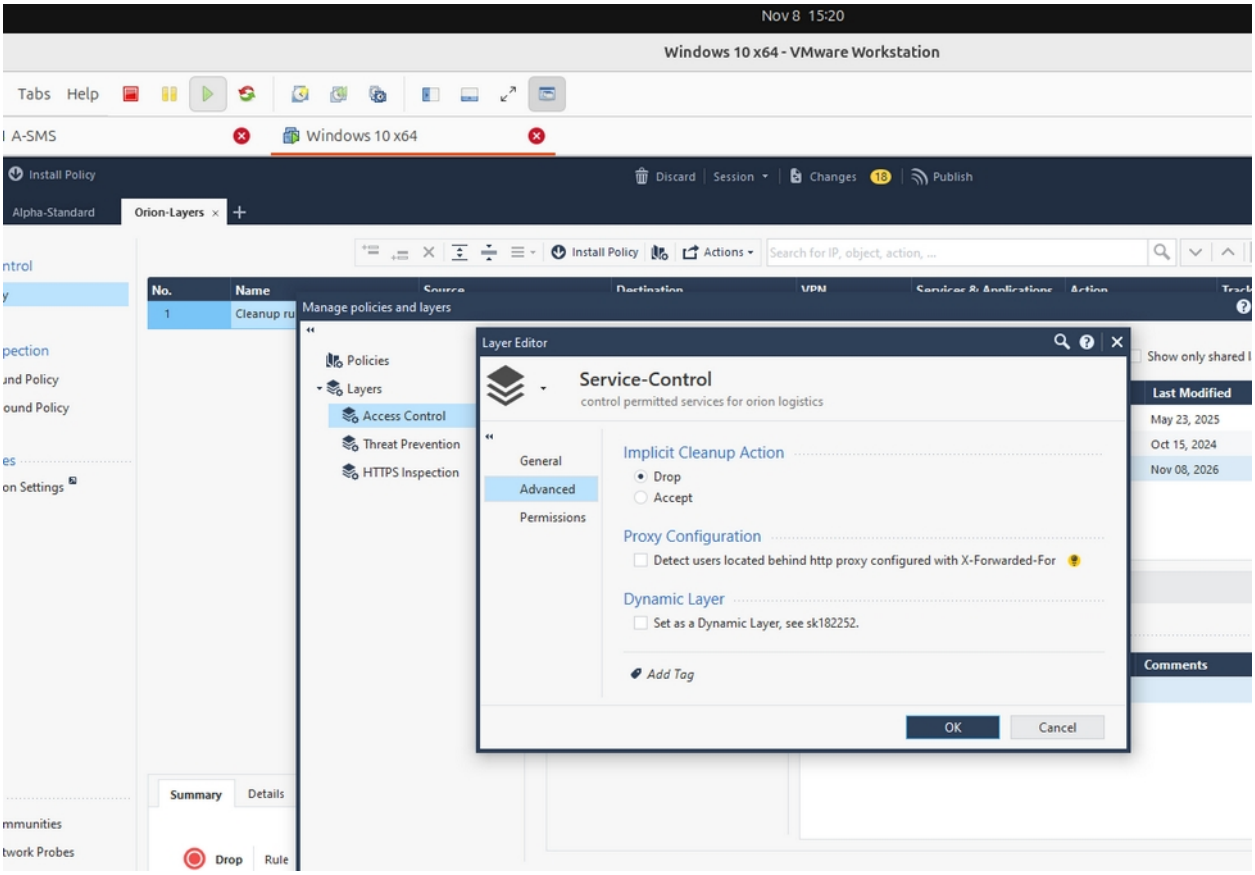


Figure C2: Advanced configuration of the Service-Control layer, showing Drop as the Implicit Cleanup Action, with Proxy Configuration and Dynamic Layer disabled.

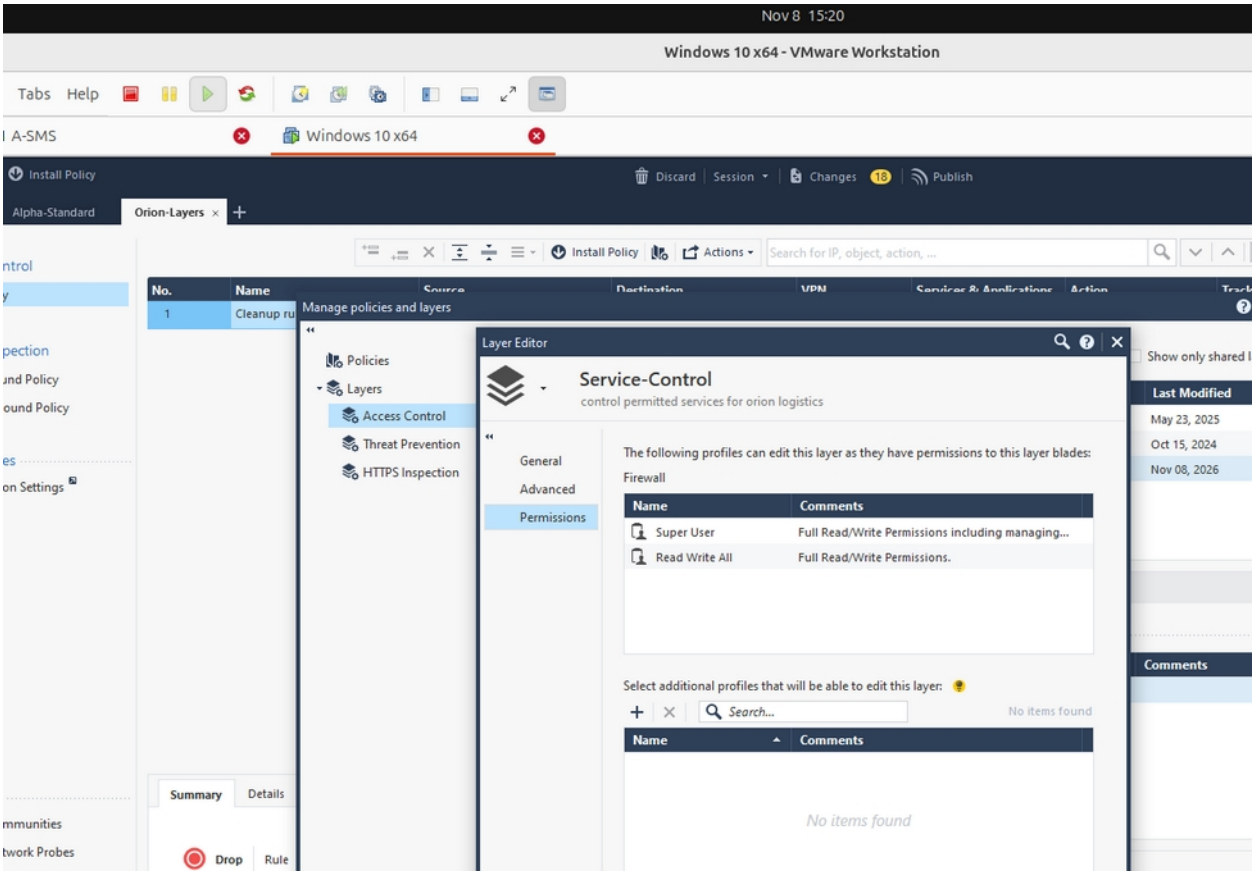


Figure C3: Permissions configuration of the Service-Control layer, retaining Super User and Read Write All as authorised editing profiles.

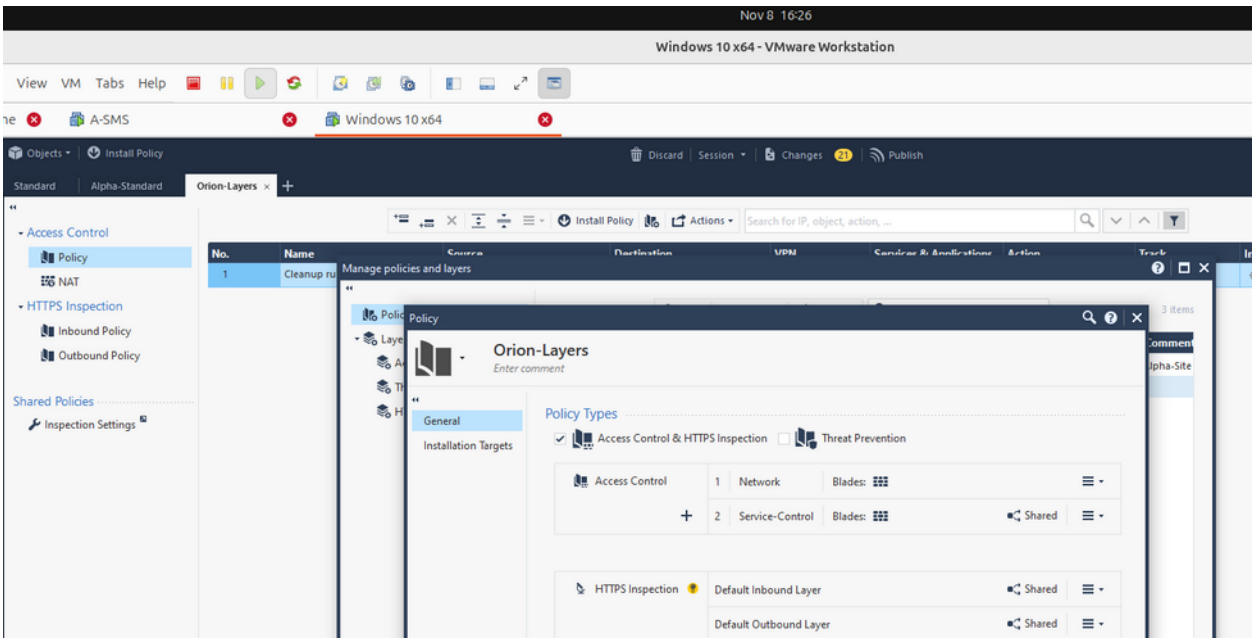


Figure C4: Orion-Layers policy package showing Network and Service-Control as two Ordered Layers, with Service-Control configured as a shared layer.

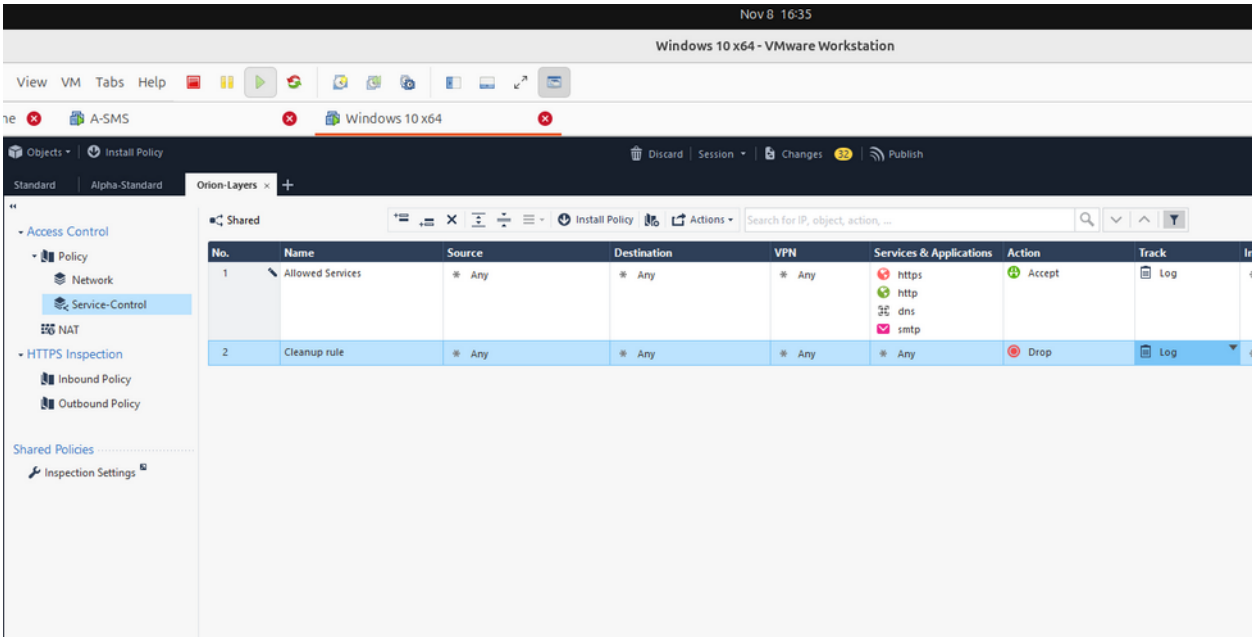


Figure C5: Service-Control Ordered Layer Rulebase

The Service-Control Ordered Layer contains an Allowed Services rule permitting HTTP, HTTPS, DNS and SMTP traffic with logging enabled. A Cleanup Rule follows, configured to drop all unmatched traffic and record the events in the logs.

5. Task 4 – Install, test and fix

Task 4 requires installation of Orion-Layers on the Security Gateway, an initial SSH test (T1), inspection of the matching log entry, and then a corrective Admin Services rule at position 1 of Service-Control. The assignment then requires publishing and installing again and running tests T1–T5.

The policy configuration was published in SmartConsole. However, the Security Gateway VM was not running because the host computer could not support running it alongside A-SMS. Therefore, installation success, the initial T1 log, the Admin Services correction and the T1–T5 traffic logs have not been demonstrated.

Required item	Status / note
Initial policy installation	Pending – Security Gateway unavailable
Initial T1 SSH test and log	Pending – no live gateway log
Admin Services rule at Service-Control position 1	Not yet added; must be completed when returning to Task 4
Second publish and installation	Pending
T1–T5 logs	Pending

Expected explanation to complete after observing the initial T1 log: an Accept in the Network layer does not by itself guarantee that the connection is finally accepted, because the packet is evaluated by the next

Ordered Layer as well. The Service-Control layer must also contain a matching permit rule; its cleanup behaviour can otherwise prevent the connection.

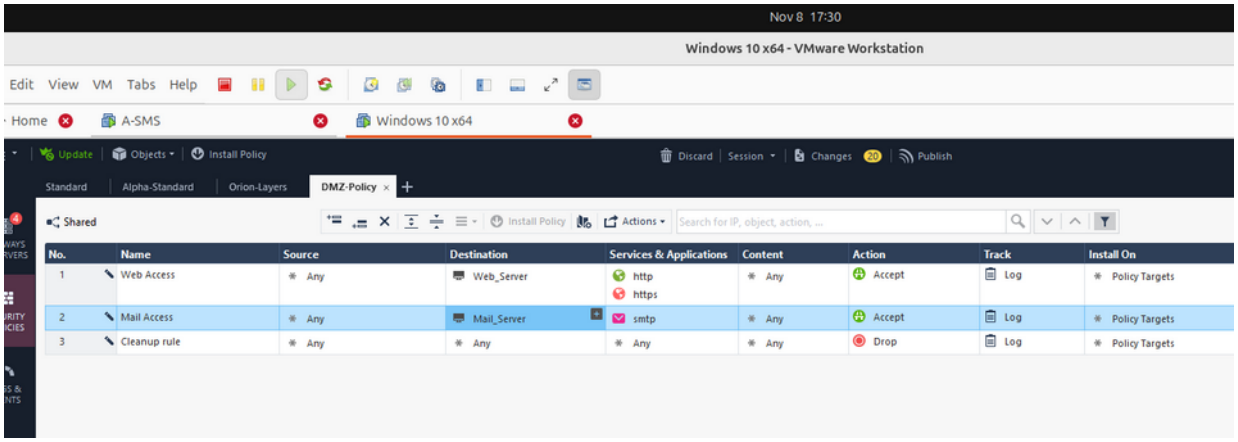
PENDING PRACTICAL VERIFICATION: successful installation evidence, initial T1 log, Admin Services rule screenshot, and individual T1–T5 log screenshots after a gateway is available.

6. Task 5 – Create DMZ-Policy Inline Layer

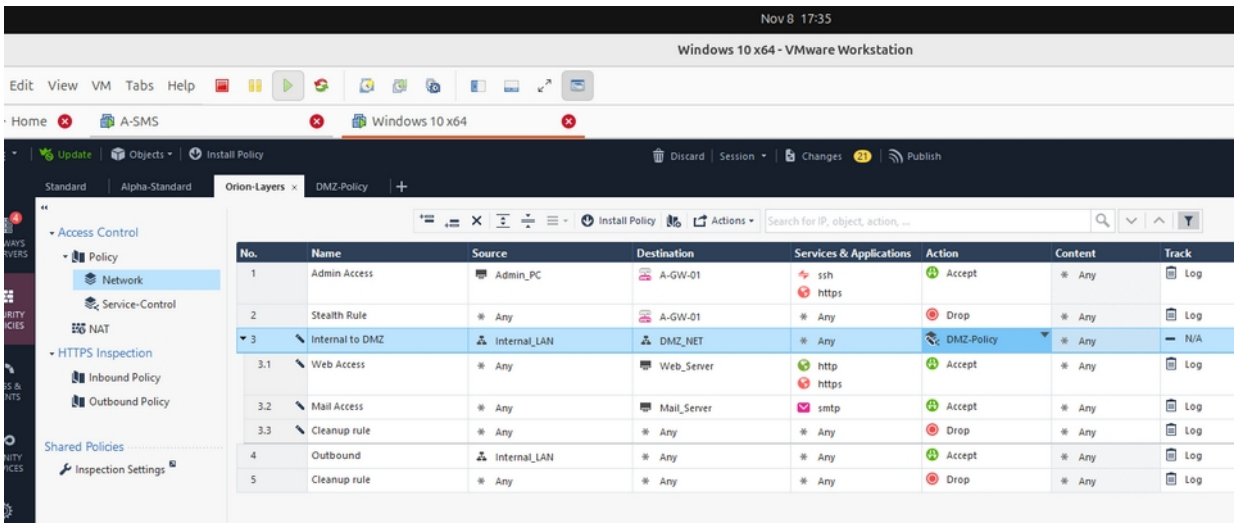
The DMZ-Policy layer was created and configured with Firewall, Applications & URL Filtering and Content Awareness enabled, Mobile Access disabled, Sharing enabled, Implicit Cleanup Action Drop, and edit permissions for Super User and Read Write All.

No.	Name	Source	Destination	Services	Action	Track
1	Web Access	Any	Web_Server	HTTP, HTTPS	Accept	Log
2	Mail Access	Any	Mail_Server	SMTP	Accept	Log
3	Cleanup Rule	Any	Any	Any	Drop	Log

The action of Network Rule 3 (Internal to DMZ) was changed from Accept to the DMZ-Policy Inline Layer. The nested Web Access, Mail Access and Cleanup Rule entries appeared beneath Rule 3. The changes were published.



Screenshot T5-A: DMZ-Policy rulebase showing Web Access, Mail Access and Cleanup Rule.



Screenshot T5-B: Network layer showing DMZ-Policy attached to Rule 3.

7. Task 6 – Install and verify Inline Layer

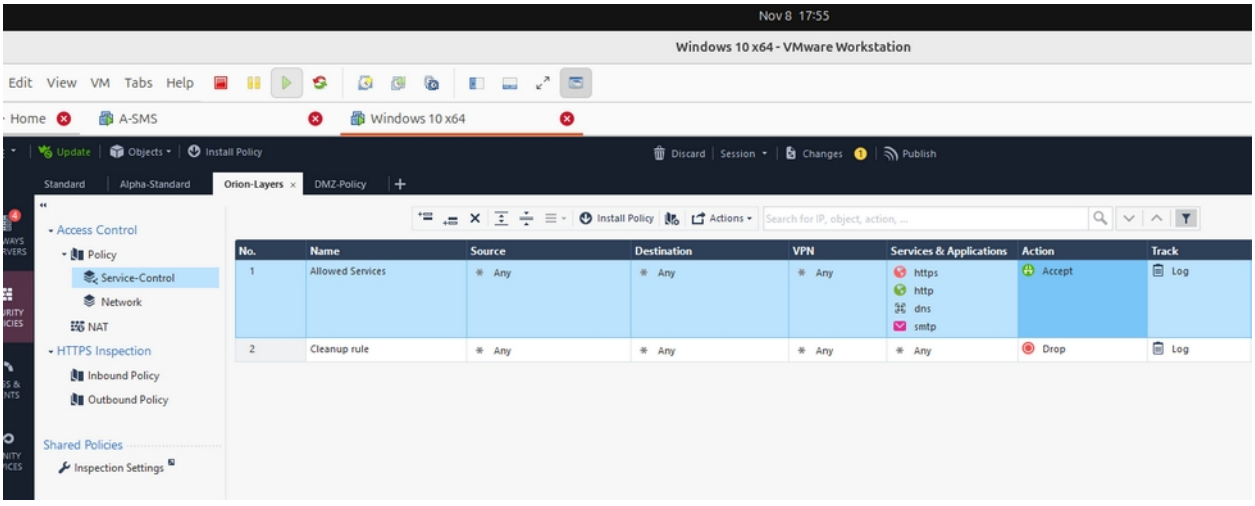
Task 6 requires successful installation of Orion-Layers and tests T6–T9. It also requires comparison of the T5 and T8 logs. Because the Security Gateway was unavailable, these live tests and log comparisons remain outstanding.

Test	Required traffic	Result	Matching layer/rule
T6	Internal host browses to Web_Server using HTTP	Pending	Pending
T7	Internal host connects to Mail_Server using SMTP (TCP 25)	Pending	Pending
T8	Internal host connects to Web_Server using Telnet (TCP 23)	Pending	Pending
T9	Internal host browses to an external server using HTTP	Pending	Pending

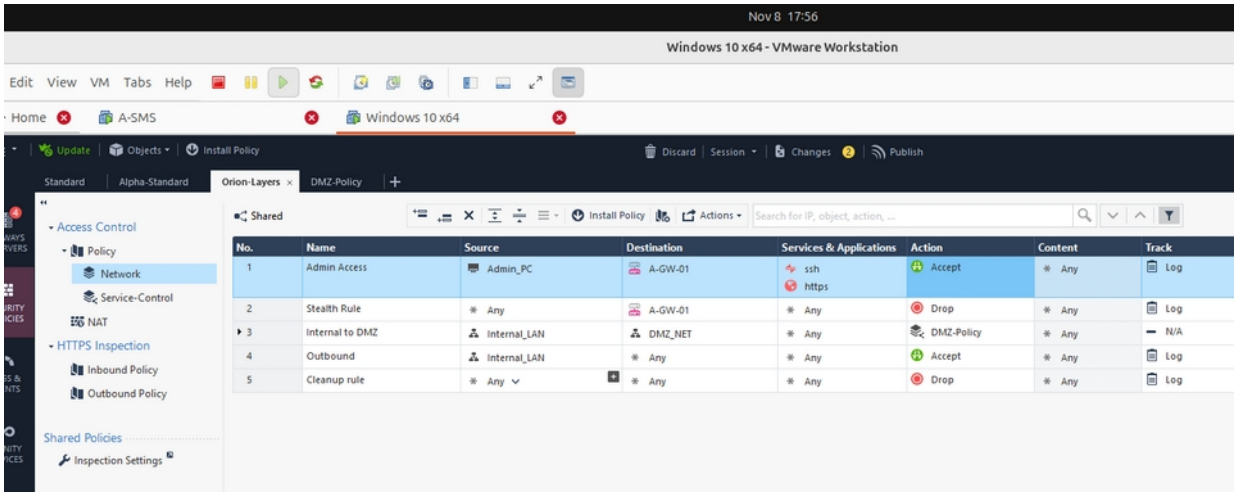
PENDING PRACTICAL VERIFICATION: successful installation screenshot and one log screenshot for each of T6–T9. Compare the actual T5 and T8 log entries only after they have been captured.

8. Task 7 – Layer order and sharing

The Access Control layer order was changed so that Service-Control appeared above Network. The order was subsequently restored to the original arrangement, with Network first and Service-Control second. The original rulebase and DMZ-Policy attachment remained present.



Screenshot T7-A: Policy properties showing Service-Control above Network during the order experiment.



Screenshot T7-B: Network rulebase after the original order was restored. Obtain a separate policy-properties screenshot showing Network first and Service-Control second if required as explicit order evidence.

Layer-order explanation: Ordered Layers are evaluated in sequence. When the order changes, the first layer that evaluates the traffic changes, so the matching layer and rule recorded in the logs may differ. If the later layer still has no matching permit rule and its cleanup action is Drop, the user-facing outcome remains a dropped connection even though the layer responsible for the drop changes.

Sharing explanation: Sharing: Enabled allows a layer to be reused by another policy package instead of rebuilding the same rulebase. Reuse can help maintain consistent controls and reduce duplicated configuration. The DMZ-Policy layer was configured with Sharing enabled.

Task 7 evidence	Status
Screenshot showing Service-Control above Network	Captured
T3 log after reorder	Pending – gateway-dependent
T4 log after reorder	Pending – gateway-dependent
Screenshot showing restored original order	Captured
Written explanation of order and sharing	Included

PENDING PRACTICAL VERIFICATION: T3 and T4 logs

9. Conclusion

The Orion-Layers policy package, Network and Service-Control Ordered Layers, and DMZ-Policy Inline Layer were configured in SmartConsole and published. The DMZ-Policy layer was attached to Network Rule 3, and the Task 7 layer-order experiment was performed before restoring the original order. Policy installation and traffic-based verification remain incomplete because the Security Gateway could not be run with the available host resources. The report therefore separates verified configuration work from pending operational evidence.

10. Part B – Theory questions

Q1. Which statement best describes a policy layer?

B. A set of rules that divides a policy into smaller, more manageable sections.

Q2. What are the names of the default Access Control policy and its default Ordered Layer?

C. Standard and Network.

Q3. A packet matches an Accept rule in the first Ordered Layer. What does the gateway do next?

A. It continues checking the next Ordered Layer. An Accept in one Ordered Layer is not the final decision for the complete layered policy.

Q4. Which type of layer is attached to a specific rule as a sub-policy?

D. Inline Layer.

Q5. What is the recommended practice at the end of each Ordered Layer?

B. Add an explicit Cleanup rule with the same Action as the Implicit Cleanup Rule.

Q6. Define a policy layer and state why layers are useful in a large network. Name two benefits.

A policy layer is a separate rulebase that forms part of an Access Control policy. Layers divide a large policy into smaller areas that are easier to review and manage. Two benefits are improved organisation and the ability to manage or reuse specific rule groups more consistently.

Q7. Compare Ordered Layers and Inline Layers. For each, say how it is evaluated and give one use case.

Ordered Layers are evaluated sequentially as part of the policy. They are useful for separating broad network controls from service-specific controls. An Inline Layer is attached to a particular rule and is evaluated when traffic matches that parent rule; it is useful for applying a dedicated set of controls to DMZ access.

Q8. Describe the order of enforcement for Ordered Layers. Include what happens when a rule action is Drop, when it is Accept, and when no rule matches.

The gateway evaluates the Ordered Layers in their configured sequence. A Drop denies the traffic. An Accept in one layer allows evaluation to continue to the next Ordered Layer rather than ending the entire policy decision. If no rule matches in a layer, its cleanup behaviour applies; with Implicit Cleanup Action Drop, the traffic is dropped.

Q9. Name the three groups of settings you configure for a layer. Give one example of a setting from each group.

General: enabled blades or Sharing. Advanced: Implicit Cleanup Action. Permissions: which permission profiles can edit the layer.

Q10. Explain what Sharing does for a layer. State one guideline from the module about sharing or reusing layers.

Sharing makes a layer available for reuse in other policy packages. A useful guideline is to share a layer when the same controlled rulebase needs to be reused, rather than maintaining duplicate copies; keep sharing disabled when the layer is intended only for its current policy.

Q11. A policy has two Ordered Layers. A connection is accepted by a rule in the first layer. The second layer has no rule that matches the connection and no explicit Cleanup rule, and its Implicit Cleanup Action is Drop. What happens? Explain confirmation, fix and explicit Cleanup.

The connection is dropped by the second layer's implicit cleanup action. Confirm this by inspecting the gateway log and identifying the layer and rule or cleanup action responsible. If the connection should be allowed, add a specific matching Accept rule in the second layer before its cleanup. An explicit Cleanup rule makes the intended final action visible in the rulebase and easier to audit, even when it matches the implicit cleanup action.

Q12. A company policy has grown to hundreds of rules. Administrators find it slow to review and are afraid to change it. Propose a layer design using Ordered and Inline Layers. Explain manageability, role-based administration and performance.

Use Ordered Layers to separate broad functions—for example, Network controls followed by Service-Control rules. Use an Inline Layer for a focused area such as DMZ access, attached to the relevant parent rule. This makes each rulebase smaller and easier to review, allows permissions to be assigned to administrators responsible for particular layers, and supports reuse where appropriate. Performance should be assessed with actual policy installation and traffic testing; layers should be designed clearly and avoid unnecessary duplication or overly complex evaluation.

11. Appendix – Evidence register

Evidence ID	Description	Status
T1-A to T1-C	Default Network layer General, Advanced and Permissions screenshots	Included
T2-A	Network layer rulebase	Included
T2-B	Orion-Layers package view	Included
T3-A to T3-C	Service-Control settings pages	Included
T3-D	Orion-Layers ordered layer list	Included
T3-E	Service-Control rulebase	Included
T4-A to T4-D	Installation, initial T1, Admin Services and T1–T5 logs	Pending gateway
T5-A	DMZ-Policy rulebase	Included
T5-B	Network Rule 3 with Inline Layer attached	Included
T6-A to T6-F	Installation, T6–T9 logs and T5/T8 comparison	Pending gateway
T7-A	Service-Control above Network	Included
T7-B	Restored Network then Service-Control order	Included
T7-C/T7-D	T3 and T4 logs	Pending gateway